Brian Meng

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EDUCATION

University of Pennsylvania, School of Engineering and Applied Science | Philadelphia, PA *Expected May 2028*
B.S.E. in Computer Science; Second Major in Mathematical Economics
GPA: 3.97/4.00 | Benjamin Franklin Scholar
Relevant Coursework: Data Structures & Algorithms, Computer Systems, Computational Complexity, Big Data Analytics
Honors: Wharton Undergraduate Finance Club Market-Making Competition 2nd Place | Four-time AIME Qualifier

PROFESSIONAL EXPERIENCE

Lunar Energy | *Software Engineering Intern* *May 2026 – August 2026*
- Enabled real-time consumption forecasts for previously skipped **cold-start EV charging sessions** by building a Python charging-duration fallback in Gridshare's AWS Lambda pipeline.
- Increased detected EV charging events by **~50% in production** by reprocessing detection windows to catch multiple charging events per window, recovering events the pipeline previously missed.
- Improved battery-state estimates by adding an MAE threshold rejecting poor curve fits, **cutting error variance by 25%**, enabling more accurate charge/discharge and grid decisions in cost optimization.

Lilia AI | *ML Engineering Intern* *May 2025 – August 2025*
- Built an end-to-end Python pipeline automating DICOM ingestion and preprocessing across **200+** CT scans, producing model-ready inputs for a PyTorch-based medical model training.
- Developed and integrated a reusable Python sizing agent into a multi-agent medical-imaging system, automating millimeter-scale volume, surface area, and diameter measurements with **~3% mean relative error**.

Forkast | *Software Engineering Intern* *March 2025 – May 2025*
- Built a cross-platform **React Native** mobile prototype of the existing restaurant dynamic-pricing app, porting onboarding, account creation, and category-level pricing controls, establishing foundation for mobile development.

PROJECTS

Distributed Key-Value Store | Go, Docker, AWS EC2 *Summer 2026*
- Designed a partitioned, replicated key-value store in Go, enabling durable storage and continued reads during failures with consistent hashing, leader-based replication, and gossip-based failure detection.
- Cut request recovery after leader failure from **up to 3 seconds to approximately 10 ms** using immediate replica fallback; containerized the multi-node cluster and automated testing with GitHub Actions.

Distributed Job Scheduler | Go, PostgreSQL, Docker *Summer 2026*
- Built a multi-process job scheduler in Go/PostgreSQL that delegates tasks to independent workers and guarantees each runs at least once, using atomic job claiming, fencing tokens, failure detection, and exponential retry backoff.
- Recovered jobs from crashed workers in **~48 ms,** instead of indefinitely, via heartbeat timeouts and auto-requeuing.
- Improved concurrent job claiming throughput **2.8x** by utilizing row-level locking and increasing worker processes.

Career@Penn | Next.js, TypeScript *Spring 2026*
- Led a **six-person** Penn Spark team developing a career info platform for Penn students, launched on Vercel.
- Completed core authentication and profile functionality via Google OAuth, cookie-based sessions, guest-profile persistence, validated by a 60-test Vitest suite.

TECHNICAL SKILLS

Languages: Java, Python, Go, TypeScript/JavaScript, C/C++, SQL, R, OCaml
Frameworks: Next.js, React, React Native, Flask, SQLAlchemy
Cloud/Data: AWS (Lambda, EC2, S3, Redshift, CloudWatch), Docker, PostgreSQL, SQLite, Elasticsearch, Linux/Unix

LEADERSHIP & ACTIVITIES

Consult for America | *Project Lead, Consultant* *January 2025 – Present*
- Led teams across two client engagements, delivering go-to-market strategies and clean-energy regulatory analysis.

Activities: The Mask and Wig Club (Assistant Stage Manager), Upstage A Cappella, Wharton Asia Exchange
Spoken Languages: Mandarin Chinese (proficient), French (conversational) | **Interests:** Poker, music production, theatre